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Gen2-IC: Bridging Generative Models and Image Codecs through Latent Transport

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Abstract

Diffusion-based image compression has achieved strong perceptual quality at ultra-low bitrates. However, existing codecs are often tied to specific backbones and specialized components, making diverse, rapidly evolving generative models difficult to reuse. This raises a natural question: *Can modern generative foundation models be connected to image compression through a simple and extensible interface?* Two insights guide our design: stronger generative priors make a simpler codec interface viable, and generation and compression can be intrinsically linked through latent transport. We therefore propose Gen2-IC with two stages: (1) *Latent Compression* maps clean image latents to entropy-constrained latents; and (2) *Latent Transport* refines them with one near-terminal update based on the pretrained model. Gen2-IC requires neither auxiliary conditioning signals nor task-specific backbone modifications. With lightweight adaptation and no distillation, it supports fast encoding and one-step decoding across multiple bitrates. We validate Gen2-IC on SD-2.1, SANA-1.5, FLUX.1-dev, and Qwen-Image-2512, spanning U-Net and Transformer architectures as well as diffusion and flow-matching formulations. With stronger priors, Gen2-IC delivers gains below 0.05 bpp: the Qwen variant leads diffusion-based generative codecs in reconstruction fidelity (PSNR), perceptual similarity (LPIPS and DISTS), and recognizer-based semantic fidelity (OCR CER/WER and face-ROI similarity) across four benchmarks.

1 Introduction

Lossy image compression aims to reduce bitrate while preserving reconstruction fidelity and perceptual quality. At ultra-low bitrates, the severe information bottleneck intensifies the perception–distortion trade-off (Blau and Michaeli 2018, 2019). Under this constraint, distortion-oriented objectives (e.g., MSE) tend to average uncertain details, causing blur and texture loss. Perceptual losses (e.g., LPIPS (Zhang et al. 2018) and DISTS (Ding et al. 2020)) and adversarial training (Goodfellow et al. 2014) improve visual realism (Yan et al. 2021; Mentzer et al. 2020; Muckley et al. 2023), but remain insufficient to preserve semantic fidelity, leading to errors such as corrupted text or facial identity drift.

Modern generative foundation models are rapidly improving their ability to model text, faces, objects, and fine visual structure (Rombach et al. 2022; Xie et al. 2025; Black Forest Labs 2024; Wu et al. 2025). These models provide increasingly powerful generative priors for recovering details omitted by a limited bit budget. However, a core challenge remains: the objectives of generation and compression are not naturally aligned. Generation favors diverse, plausible outputs, whereas compression requires faithful reconstruction of a specific source without semantic drift.

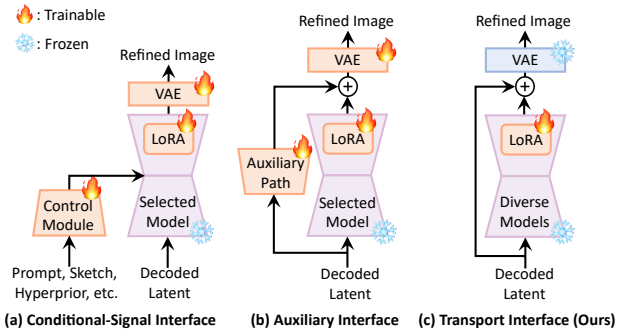

Figure 1: Generative-codec interfaces. Existing methods formulate generative compression as conditional generation, coupling a selected model with conditioning signals or auxiliary paths. Gen2-IC instead formulates reconstruction as latent transport and directly refines the decoded latent through a single path, enabling lightweight adaptation and one-step, multi-rate decoding across diverse models.

As illustrated in Figure 1, existing generative codecs bridge this gap by casting reconstruction as conditional generation and coupling a selected generative model with conditioning signals or auxiliary modules (Lei et al. 2023; Careil et al. 2023; Zhang et al. 2025; Xue et al. 2025b; Shi et al. 2026). To reduce decoding latency, one-step variants further use distillation objectives or rate–timestep mappings (Zhang et al. 2025; Xue et al. 2025b; Guo et al. 2025; Shi et al. 2026). Although effective, these mechanisms often remain tightly tied to particular backbones and native sampling processes. Consequently, extending these methods to new generative models across diverse architectures and generative processes requires model-specific codec redesign, while adapting larger models raises the costs, leaving their scaling potential underexplored in image compression. This raises a central question: *Can modern generative foundation models be connected to image compression through a simple and extensible interface?*

To answer this, we move beyond model-specific condition-

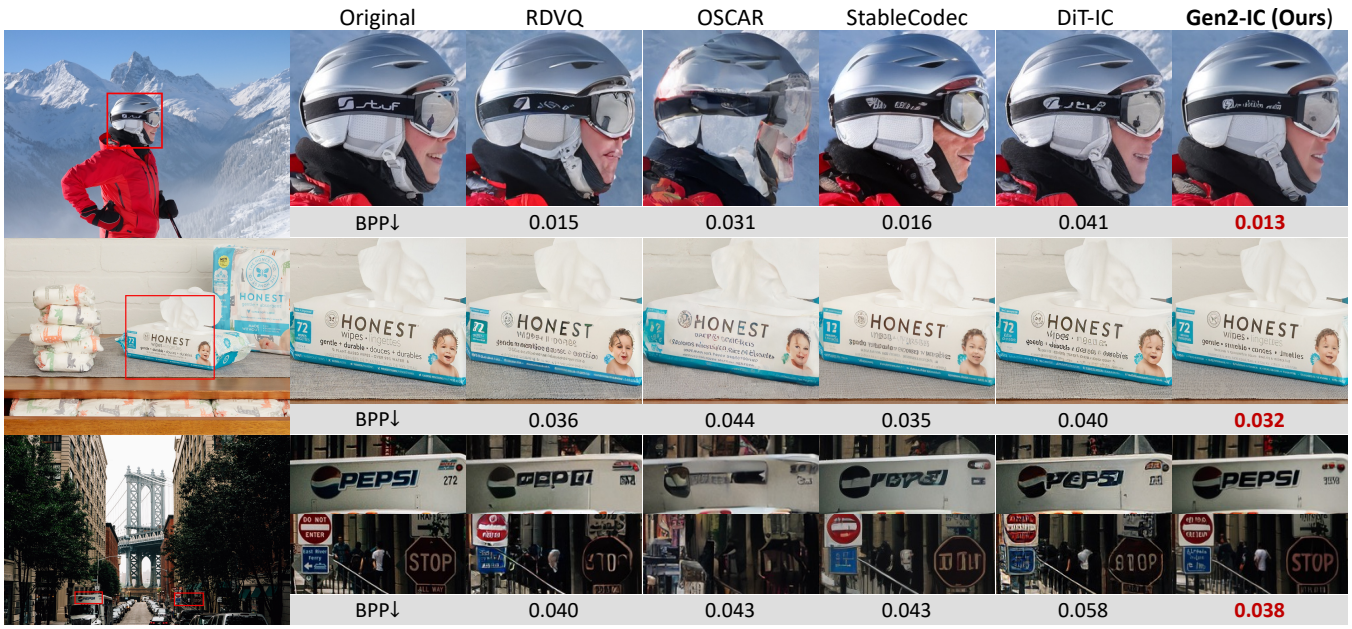

Figure 2: Source fidelity at ultra-low bitrates. Compared with RDVQ (Jiang et al. 2026), OSCAR (Guo et al. 2025), StableCodec (Zhang et al. 2025), and DiT-IC (Shi et al. 2026), Gen2-IC-Qwen better preserves text and facial details while using the lowest bitrate, with the clearest gains in semantically sensitive regions where small errors alter meaning or identity.

ing and revisit the intrinsic relationship between generation and compression. This perspective yields two key insights: (1) by better compensating for information loss at ultra-low bitrates, a stronger decoder-side prior can maintain reconstruction quality with a direct, single-path codec and fewer model-specific components; and (2) generation and compression can be coupled at the mechanism level. Specifically, a generative process transports noise toward the data distribution in latent space, whereas compression maps source data to an entropy-constrained latent distribution. This suggests a latent-transport view for integrating generative models into image compression: the prior transports decoded latents back toward the clean latent distribution before VAE synthesis.

These insights lead to **Gen2-IC**, an effective, efficient, and extensible framework that decouples image coding from generative refinement within a single reconstruction path. Specifically, **Latent Compression** uses a learned codec to map clean latents to entropy-constrained latents, while **Latent Transport** refines them with one near-terminal update from a pretrained generative model. The two stages are designed and optimized separately, allowing Gen2-IC to benefit from advances in both compression and generation as these fields evolve. With lightweight backbone adaptation, it supports fast encoding and distillation-free one-step decoding across multiple bitrates, without additional conditioning or model-specific backbone redesign.

As previewed in Figure 2, Gen2-IC preserves text and facial details more faithfully while using the lowest bitrate in each comparison. Extensive experiments further show that the same formulation remains effective across diverse generative models, while stronger priors yield additional perceptual and semantic gains. This perspective offers a practical route for image compression to benefit directly from advances in generative modeling. Our contributions are as follows:

1. We formulate the integration of generative models into image compression as latent transport, shifting the focus from codec design for a single generator to a reusable interface across evolving generative-model families.
2. We introduce Gen2-IC, a decoupled design in which Latent Compression and Latent Transport are optimized separately, with lightweight backbone adaptation. It supports fast encoding, one-step decoding, and multi-rate operation without additional conditioning, prior-specific backbone redesign, or distillation.
3. We validate the same formulation across SD-2.1, SANA-1.5, FLUX.1-dev, and Qwen-Image-2512. A matched SD-2.1 comparison against OSCAR demonstrates quality and efficiency gains independent of model scale, while the larger FLUX and Qwen variants lead in perceptual quality and recognizer-based text and face preservation.

2 Related Work

Learned and Perceptual Image Compression. Learned codecs follow the transform-coding pipeline of traditional codecs (Wallace 1992; VTM-23.10 2025), replacing handcrafted components with neural networks for end-to-end nonlinear transforms and entropy estimation, often strengthened by hyperpriors and context models (Ballé et al. 2018; He et al. 2022; Liu, Sun, and Katto 2023; Jia et al. 2025). This paradigm has delivered strong rate–distortion performance and remains computationally practical, but distortion-

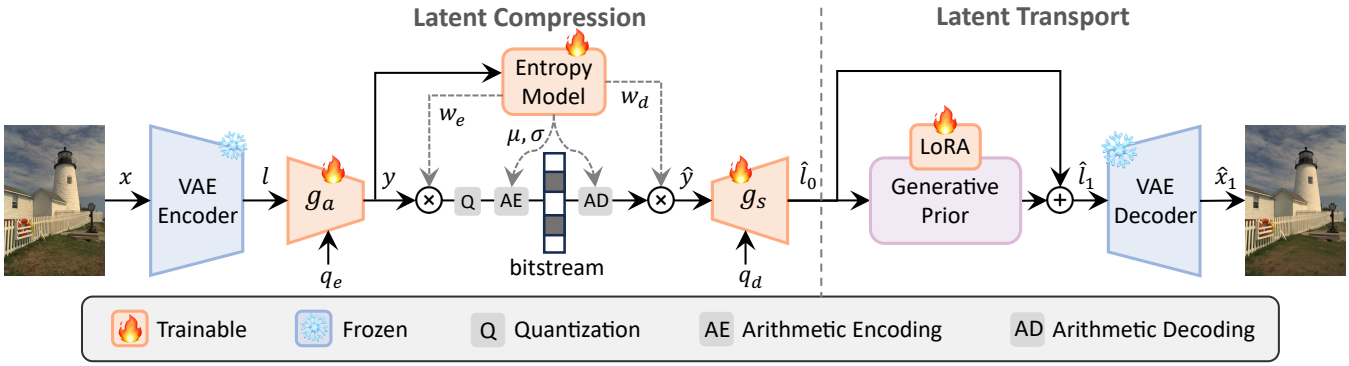

Figure 3: Overview of Gen2-IC. Latent Compression encodes the clean VAE latent l into a bitstream and reconstructs an entropy-constrained latent $\hat{l}_0$. Latent Transport refines $\hat{l}_0$ through one near-terminal update before VAE decoding. The two stages are optimized separately, allowing the same formulation to be instantiated across diverse generative models.

oriented training often yields blur and texture loss at low rates. Perceptual codecs therefore add learned perceptual losses and adversarial training (Yan et al. 2021; Mentzer et al. 2020; Muckley et al. 2023), while codebook-based methods exploit discrete representations and nonlinear modeling capacity for perceptual gains at high compression ratios (Jia et al. 2024; Xue et al. 2025a; Jiang et al. 2026). Nevertheless, at ultra-low bitrates, preserving fine-grained semantics and structure, including facial identity, geometry, and text, remains challenging. Curated data and structure-specific priors can help (Jia et al. 2024; Wang, Mallya, and Liu 2021; Tang et al. 2026), but add engineering cost.

Generative Priors for Image Compression. Diffusion-based codecs exploit generative priors to recover details that rate-constrained representations cannot retain. Existing methods commonly cast reconstruction as conditional generation, guiding a selected generative model with prompts, sketches, or hyperprior features (Lei et al. 2023; Careil et al. 2023; Xue et al. 2025b). Many rely on iterative or few-step sampling (Yang and Mandt 2023; Li et al. 2025a,b; Ke et al. 2025). Recent methods reduce decoding latency through one-step reconstruction, using distillation or rate- and variance-aware timestep alignment (Zhang et al. 2025; Xue et al. 2025b; Guo et al. 2025; Shi et al. 2026); some additionally introduce auxiliary paths or semantic distillation to improve reconstruction (Zhang et al. 2025; Xue et al. 2025b; Shi et al. 2026). Although effective, these interfaces are usually developed around a particular backbone and sampling process. Meanwhile, generative models are rapidly evolving in both formulation and architecture, from denoising diffusion (Ho, Jain, and Abbeel 2020; Rombach et al. 2022) to flow matching (Lipman et al. 2023) and rectified flow (Liu, Gong, and Liu 2023), and from U-Nets to scalable Transformers (Esser et al. 2024; Xie et al. 2025; Black Forest Labs 2024; Wu et al. 2025; Cao et al. 2025; Gao et al. 2025). Trained on large-scale image–text corpora, they provide increasingly strong semantic and visual priors. Recent work has also explored diffusion foundation-model pretraining specifically for image compression (Jia et al. 2026). Yet the growing diversity of generative models makes model-specific integration difficult to scale. Gen2-IC instead provides a simple and extensible latent-transport interface that connects image compression to existing generative models across architectures and generative processes.

3 Method

As shown in Figure 3, Gen2-IC consists of two stages. **Latent Compression** maps the input image x through the frozen VAE encoder $\mathcal{E}_e(\cdot)$ into a clean latent l and compresses it into an entropy-constrained latent $\hat{l}_0$; **Latent Transport** then applies one near-terminal update to obtain $\hat{l}_1$, from which the frozen VAE decoder reconstructs $\hat{x}_1 = \mathcal{E}_d(\hat{l}_1)$.

Latent Compression. Latent Compression learns an efficient multi-rate codec in the VAE latent space. It reconstructs an entropy-constrained latent $\hat{l}_0$ close to the clean latent l, providing an image-specific initialization for Latent Transport. To improve parameter efficiency while preserving nonlinear modeling capacity, we adopt depthwise-convolutional transforms (Jia et al. 2025), together with a four-step quadtree autoregressive entropy model $\phi(\cdot)$ (Li, Li, and Lu 2023). The entropy model predicts Gaussian parameters (μ, σ) for coding the quantized representation $\hat{y}$, together with quantization steps (w_e, w_d). To support multiple bitrates within a single latent-codec model, learnable channel-wise scales (q_e, q_d) modulate the analysis and synthesis transforms $g_a(\cdot)$ and $g_s(\cdot)$ (Li, Li, and Lu 2024). Combined with (w_e, w_d), these scales control the operating rate. Additional details are provided in the appendix. Given an input image x, the complete latent-coding process is:

$$
\begin{aligned}
\text{Latent extraction: } & l = \mathcal{E}_e(x), \\
\text{Latent encoding: } & y = g_a(l, q_e), \\
\text{Entropy estimation: } & (\mu, \sigma, w_e, w_d) = \phi(y), \\
\text{Quantization: } & \hat{y} = \Big(\text{round}(y \odot w_e - \mu) + \mu\Big) \odot w_d, \\
\text{Latent decoding: } & \hat{l}_0 = g_s(\hat{y}, q_d).
\end{aligned} \tag{1}
$$

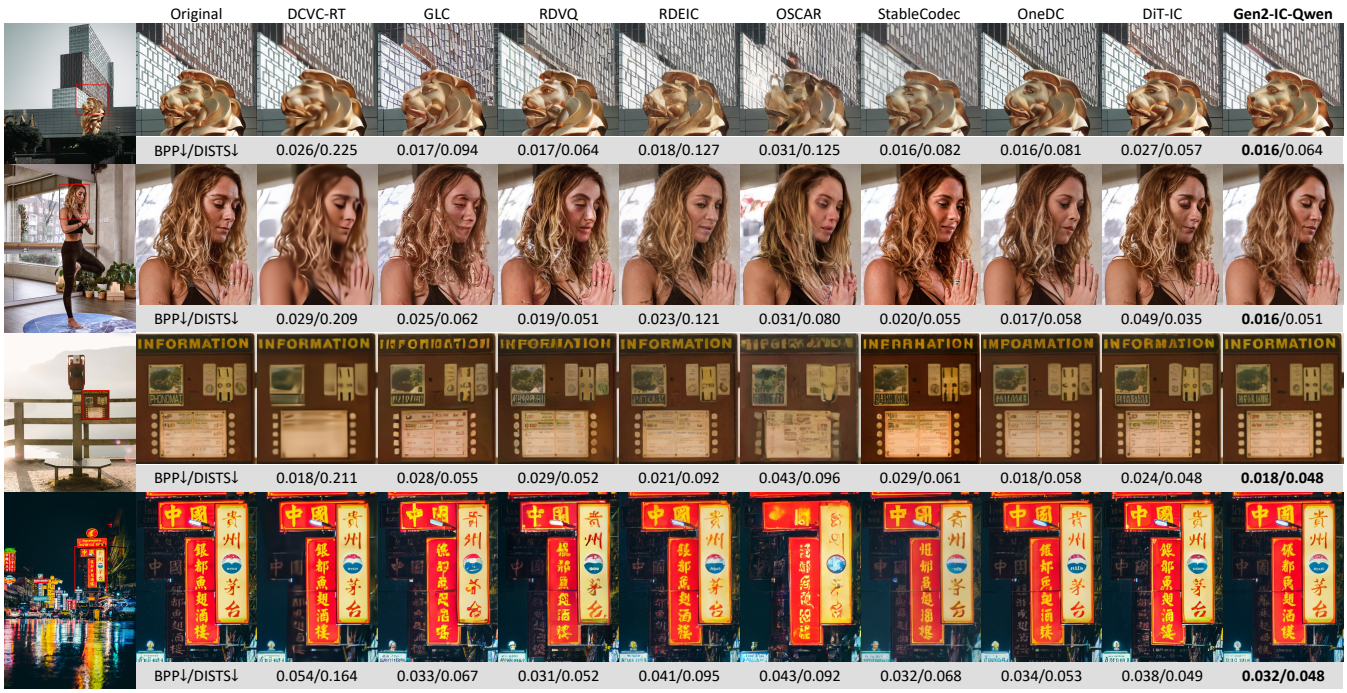

Figure 4: Qualitative comparison at ultra-low bitrates. Cropped regions highlight portraits, text, and fine structures.

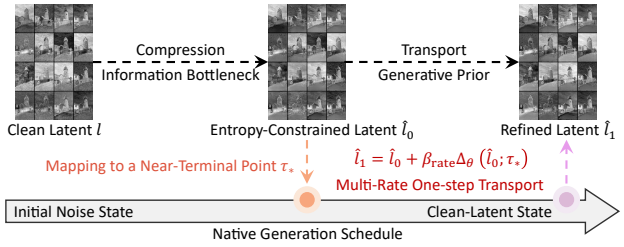

Figure 5: Rate-aware Latent Transport. Gen2-IC maps the entropy-constrained latent $\hat{l}_0$ to a near-terminal state of the generative model's native schedule and applies one update, with β_{rate} controlling the correction strength across bitrates.

Latent Transport. Latent Compression maps l to an entropy-constrained latent $\hat{l}_0$ through quantization and the information bottleneck. Although $\hat{l}_0$ retains the source content and layout, it may lose fine texture or distort structured regions. Because $\hat{l}_0$ is source-derived and already close to l, it provides a more informative initialization than the random noise used in generation. We therefore use $\hat{l}_0$ to construct a state near the end of the generative model's native schedule and apply one scheduler update to obtain $\hat{l}_1$, as illustrated in Figure 5. The decoded latent itself supplies the image-specific signal, requiring no image-specific prompt or auxiliary semantic input. The resulting update is:

$$\hat{l}_1 = \hat{l}_0 + \beta_{\text{rate}}\,\Delta_\theta\big(\hat{l}_0;\tau_\star\big), \tag{2}$$

where $\Delta_\theta(\hat{l}_0;\tau_\star)$ denotes the correction induced by one native scheduler update, and $\tau_\star$ is the final update point in each model's native schedule before VAE decoding. Near-terminal placement limits the generative prior's freedom, reducing over-refinement and hallucination risk.

Unlike approaches that map each bitrate to a pseudo-timestep (Guo et al. 2025; Shi et al. 2026), Gen2-IC controls multi-rate refinement through β_{rate} at the fixed $\tau_\star$, using larger values at lower bitrates and smaller values at higher bitrates. This supports multiple rates without rate-specific adapters while preserving one-step decoding.

Decoupled Optimization. Gen2-IC optimizes Latent Compression and Latent Transport in separate stages, avoiding joint end-to-end training. The codec is trained first and then frozen while only the LoRA (Hu et al. 2022) parameters are adapted for transport. The VAE encoder $\mathcal{E}_e(\cdot)$ and decoder $\mathcal{E}_d(\cdot)$ remain frozen throughout. The latent codec comprises the analysis transform $g_a(\cdot)$, synthesis transform $g_s(\cdot)$, and entropy model $\phi(\cdot)$. We first optimize these components with a latent-space rate–distortion objective:

$$\mathcal{L}_{\text{LC}}^{(1)} = \lambda_{\text{rate}}\,\text{MSE}\Big(l,\hat{l}_0\Big) + R(y), \tag{3}$$

where $R(y)$ denotes the bitrate estimated by $\phi(\cdot)$, and λ_{rate} controls the rate–distortion trade-off. This objective bypasses the VAE decoder and accelerates convergence.

We then obtain $\hat{x}_0 = \mathcal{E}_d(\hat{l}_0)$ and refine the latent codec with an image-space objective:

$$\begin{aligned}\mathcal{L}_{\text{LC}}^{(2)} = \lambda_{\text{rate}}\Big(&\mathcal{L}_1(x,\hat{x}_0) + \mathcal{L}_{\text{per}}(x,\hat{x}_0)\\ &+ \mathcal{L}_{\text{adv}}(x,\hat{x}_0)\Big) + R(y),\end{aligned} \tag{4}$$

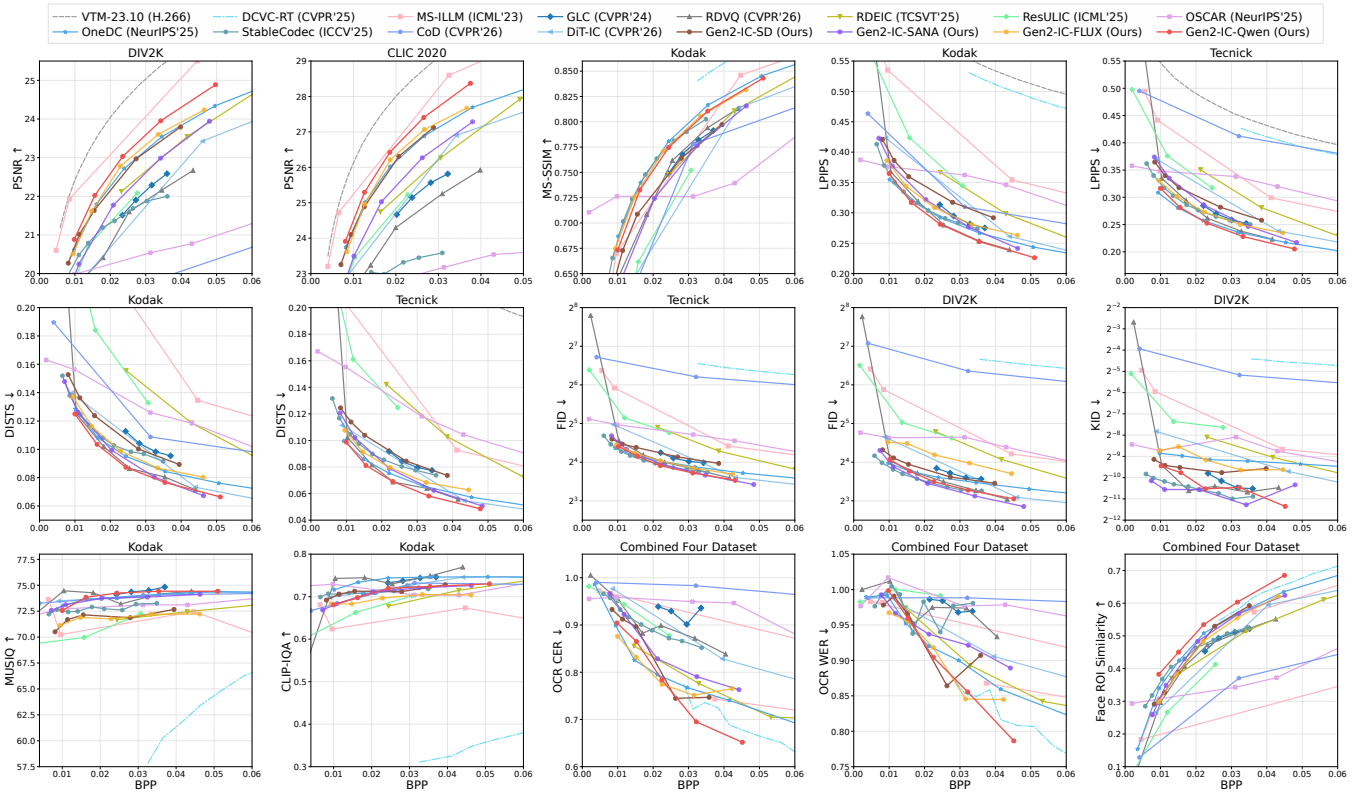

Figure 6: Rate–quality comparison on representative dataset–metric pairs. Higher is better for PSNR, MS-SSIM, MUSIQ, CLIP-IQA, and ROI similarity, while lower is better for LPIPS, DISTS, FID, KID, CER, and WER. CER, WER, and ROI similarity are evaluated over the combined Kodak, Tecnick, DIV2K, and CLIC 2020 datasets.

where $\mathcal{L}_1(\cdot)$, $\mathcal{L}_{\text{per}}(\cdot)$ (Zhang et al. 2018), and $\mathcal{L}_{\text{adv}}(\cdot)$ (Goodfellow et al. 2014) denote the L1 reconstruction, perceptual, and adversarial losses, respectively. This objective directly improves decoded image $\hat{x}_0$ quality.

After optimizing Latent Compression, we freeze the latent codec and adapt only lightweight LoRA parameters of the pretrained generative backbone. The objective is:

$$\mathcal{L}_{\text{LT}} = \mathcal{L}_1(x, \hat{x}_1) + \mathcal{L}_{\text{per}}(x, \hat{x}_1) + \mathcal{L}_{\text{adv}}(x, \hat{x}_1), \quad (5)$$

where $\hat{x}_1 = \mathcal{E}_d(\hat{l}_1)$ is decoded from the refined latent $\hat{l}_1$. This source-referenced loss drives the LoRA-adapted prior to learn a residual correction at the source-anchored near-terminal state rather than perform unconstrained generation.

4 Experiments

4.1 Experimental Setup

Training. We instantiate Gen2-IC with SD-2.1 (Rombach et al. 2022), SANA-1.5 (Xie et al. 2025), FLUX.1-dev (Black Forest Labs 2024), and Qwen-Image-2512 (Wu et al. 2025), using training and validation subsets from ImageNet (Deng et al. 2009) and OpenImages (Kuznetsova et al. 2020). Each variant uses its native VAE, latent codec, scheduler conversion, and LoRA adapter; the two-stage formulation is shared. Latent Compression and Latent Transport use 256×256 and 512×512 crops, while the codec supports five rates indexed by $\text{qp} \in \{0, \ldots, 4\}$. For transport, the codec, VAE, and backbone are frozen; one LoRA adapter per model ($r = 16$, $\alpha = 32$, dropout 0.0) is trained at $\text{qp} = 0$ and reused across rates. The specific β_{rate} values are selected on the validation set and fixed during testing, without per-image or per-dataset adjustment (e.g., $\{0.99, 0.85, 0.55, 0.45, 0.30\}$ for Gen2-IC-Qwen). Model-specific hyperparameter selection is provided in the appendix.

Evaluation. We evaluate on Kodak (Kodak 1993), Tecnick (Asuni and Giachetti 2014), DIV2K (Agustsson and Timofte 2017), and CLIC 2020 Professional (Toderici et al. 2020). We report BD-rate (Bjontegaard 2001) (PCHIP over the common metric range without extrapolation; negative indicates savings), PSNR, MS-SSIM (Wang, Simoncelli, and Bovik 2003), LPIPS (Zhang et al. 2018) with VGG features (Simonyan and Zisserman 2014), DISTS (Ding et al. 2020), MUSIQ (Ke et al. 2021), and CLIP-IQA (Wang, Chan, and Loy 2023). FID (Heusel et al. 2017) and KID (Bińkowski et al. 2018) follow the GLC protocol (Jia et al. 2024). For recognizer-based semantic evaluation, CRAFT (Baek et al. 2019) and SCRFD (Guo et al. 2022) detect source text and face ROIs, respectively, which are reused across all reconstructions (348 text ROIs in 86 images; 215 face ROIs in 74 images). CER/WER (Levenshtein 1966) measure normalized edit distances between source and reconstructed EasyOCR transcripts using CRNN (Shi, Bai, and Yao 2017), while face-

Method	BD-rate (%) ↓												**Average**
	Kodak			Tecnick			DIV2K			CLIC 2020			
	PSNR	LPIPS	DISTS	PSNR	LPIPS	DISTS	PSNR	LPIPS	DISTS	PSNR	LPIPS	DISTS	
Distortion-oriented methods													
VTM-23.10 (Intra)	-34.11	313.84	480.92	-12.66	295.19	515.25	-18.89	285.10	543.48	-21.32	498.64	724.83	297.52
DCVC-RT (Intra)	**-56.39**	244.62	504.80	**-46.59**	237.52	586.05	**-46.22**	230.70	636.97	**-51.33**	538.81	960.16	311.59
VAE-based methods													
MS-ILLM (anchor)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
GLC	189.71	-67.89	-67.83	151.98	-57.51	-48.98	223.66	-62.38	-62.16	212.87	-61.02	-45.58	25.41
RDVQ	189.18	-68.23	-70.96	191.12	-54.92	-59.85	268.85	-61.47	-65.89	267.20	-56.56	-50.95	35.63
Diffusion-based methods													
RDEIC	72.81	-52.02	-39.56	82.75	-31.46	-3.22	108.09	-35.33	-4.34	120.70	-19.05	13.63	17.75
ResULIC	138.75	-41.10	-38.59	108.47	-41.43	-32.76	195.05	-41.51	-20.39	165.24	-35.27	-12.42	28.67
OSCAR	426.46	-39.41	-49.05	489.42	-5.39	-4.96	602.30	-14.51	-20.57	839.93	21.16	36.02	190.12
OneDC	91.36	-71.19	-76.40	50.49	-66.11	-64.69	94.83	-67.94	-70.05	75.90	-69.79	-64.72	-19.86
StableCodec	51.47	-74.45	-78.20	117.29	-65.50	-65.03	212.04	-68.08	-74.75	460.96	-65.90	-64.13	23.81
CoD	180.11	-62.22	-67.32	641.99	139.40	441.82	994.34	147.82	664.04	1666.90	339.34	1507.01	549.43
DiT-IC	147.83	-64.47	-78.04	122.51	-52.10	-66.82	176.14	-57.11	-75.98	121.72	-56.79	-69.75	3.93
Gen2-IC variants													
Gen2-IC-SD	71.82	-62.65	-72.64	67.02	-52.51	-53.72	102.48	-57.17	-66.96	75.20	-52.52	-51.31	-12.75
Gen2-IC-SANA	114.28	-69.72	-79.58	121.19	-55.59	-64.04	155.45	-63.32	-77.24	123.80	-59.28	-67.48	-1.79
Gen2-IC-FLUX	48.50	-69.78	-76.96	39.30	-62.03	-64.13	91.84	-50.64	-60.85	69.67	-42.01	-43.63	-18.39
Gen2-IC-Qwen	44.66	**-75.84**	**-81.21**	44.23	**-69.25**	**-71.18**	67.41	**-72.21**	**-79.00**	46.92	**-70.48**	**-70.66**	**-32.22**

Table 1: Rate–distortion–perception comparison. BD-rate (%; lower is better) relative to MS-ILLM on Kodak, Tecnick, DIV2K, and CLIC 2020. The Average column aggregates PSNR, LPIPS, and DISTS across all four datasets.

ROI similarity is the cosine similarity of their ArcFace embeddings (Deng et al. 2019). These source-referenced proxies complement, not replace, dedicated ground-truth OCR and identity-verification evaluations.

Baselines. We compare against representative baselines: VTM-23.10 (VTM-23.10 2025) and DCVC-RT (Jia et al. 2025) in intra mode; VAE-based codecs MS-ILLM (Muckley et al. 2023), GLC (Jia et al. 2024), and RDVQ (Jiang et al. 2026); and diffusion-based codecs RDEIC (Li et al. 2025b), ResULIC (Ke et al. 2025), OSCAR (Guo et al. 2025), StableCodec (Zhang et al. 2025), OneDC (Xue et al. 2025b), CoD (Jia et al. 2026), and DiT-IC (Shi et al. 2026).

4.2 Main Results

Rate-Quality Performance. Figure 6 compares representative curves across distortion, perceptual, distributional, and semantic metrics. The Average column of Table 1 aggregates PSNR, LPIPS, and DISTS across four datasets: Gen2-IC variants consistently balance fidelity and perceptual quality, whereas several generative codecs incur larger PSNR penalties for LPIPS/DISTS gains; Gen2-IC-Qwen ranks best at -32.22%. These gains are not scale-only: using the same SD-2.1 prior as OSCAR, Gen2-IC-SD achieves a lower BD-rate in every reported fidelity, perceptual, and semantic metric. Table 2 further shows the recognizer-based semantic average improving monotonically across SD, SANA, FLUX, and Qwen, from 5.60% to 1.58%, -15.04%, and -31.08%. Together with the fidelity/perceptual results, this trend shows that Latent Transport enables stronger priors to improve source-referenced text and face consistency across Gen2-IC variants, rather than perceptual quality through source drift.

Qualitative Comparisons. Figure 2 compares challenging text and portrait regions with recent generative codecs, while Figure 4 broadens the comparison across representative codec families and scene content. At comparable or lower bitrates, Gen2-IC-Qwen more faithfully preserves source-specific text, facial features, and local geometry; competing reconstructions more often exhibit blur, deformation, or semantic drift. Additional results are provided in the appendix.

Complexity Analysis. Table 4 reports full-model inference without VAE tiling or slicing, model sharding, or sparse-attention acceleration. We use four warm-up runs followed by ten timed runs, and include arithmetic coding in the reported latency. Using the same SD-2.1 generative model as OSCAR, Gen2-IC-SD reduces encoding/decoding latency from $65.13/163.77$ to $27.87/103.05$ ms and peak memory from 4.74 to 2.70 GiB, while improving all reported quality metrics. Because the generative backbone is used only during decoding, the FLUX and Qwen variants retain fast encoding at 21.82 and 24.05 ms, respectively. Adaptation also remains lightweight across model scales: Latent Transport requires only 18.68M and 23.59M trainable LoRA parameters, corresponding to 0.16% and 0.12% of the FLUX.1-dev and Qwen-Image-2512 backbones. Additional runtime and memory results are provided in the appendix.

4.3 Ablation Studies

We ablate prior components and injection timing. Appendix results show that activation matters more than LoRA rank and

Method	BD-rate (%) ↓ CER	WER	ROI Similarity	**Average**
VAE-based methods				
MS-ILLM (anchor)	0.00	0.00	0.00	0.00
GLC	85.98	215.54	11.93	104.48
RDVQ	28.98	42.43	11.05	27.49
Diffusion-based methods				
RDEIC (2-step)	-3.74	6.08	17.32	6.55
OSCAR	210.29	298.42	137.66	215.46
OneDC	-30.08	-10.15	-13.94	-18.06
StableCodec	0.95	-0.76	-12.75	-4.19
CoD	831.45	582.44	136.07	516.65
DiT-IC	55.74	38.85	-2.96	30.54
Gen2-IC variants				
Gen2-IC-SD	4.16	17.53	-4.89	5.60
Gen2-IC-SANA	-4.58	16.14	-6.82	1.58
Gen2-IC-FLUX	-25.24	-18.00	-1.90	-15.04
Gen2-IC-Qwen	**-36.53**	**-29.70**	**-27.01**	**-31.08**

Table 2: Recognizer-based semantic-fidelity comparison. BD-rate (%; lower is better) relative to MS-ILLM for CER, WER, and face-ROI similarity over four benchmarks.

Model variant	BD-rate (%) ↓ PSNR	MS-SSIM	LPIPS	DISTS
FLUX: VAE only	-7.35	-0.54	87.27	136.52
FLUX: VAE + Prior	-12.50	-9.21	19.13	33.08
Qwen: VAE only	15.79	13.77	57.90	78.32
Qwen: VAE + Prior	-12.93	-6.79	-4.74	7.14

Table 3: Generative-prior contribution on Kodak. BD-rate (%; lower is better) relative to the pixel-codec baseline used by the FLUX and Qwen variants.

prompts bring no consistent gain, with visual comparisons.

Generative-Prior Contribution. Table 3 isolates Latent Transport by holding the VAE and latent codec fixed. Adding the prior improves all four metrics for FLUX and Qwen, with matched SD and SANA results in the supplement, demonstrating effectiveness across distinct latent spaces and generative formulations. Despite favorable training-aligned metrics, the pixel-codec baseline shows weaker fidelity and recognizer-based semantic preservation, while supplementary user-study results show lower human preference. Using MS-ILLM as the BD-rate anchor, Gen2-IC-Qwen with Latent Transport reduces the CER, WER, and face-ROI BD-rates from $7.49/-9.33/-15.46\%$ for the pixel-codec baseline to $-36.53/-29.70/-27.01\%$, respectively. These results demonstrate that Latent Transport improves reconstruction quality and recognizer-based semantic preservation.

Near-Terminal Injection. We scan Qwen's native 50-step trajectory at $\mathrm{qp} = 0$. At each injection position, $\beta^\star$ is selected on the validation set using the frozen backbone without LoRA, and then fixed while the LoRA adapter is trained for 100 updates. Table 5 reports the resulting Kodak DISTS.

Method	S	Enc. C/T	Dec. C/T	Model P/M
Diffusion-based methods				
RDEIC	2	2.26/157.25	7.60/426.68	1.38/6.86
ResULIC	3	2.08/96.24	8.79/428.32	1.39/7.99
OSCAR	1	2.11/65.13	6.07/163.77	1.01/4.74
OneDC	1	2.62/100.50	6.46/235.03	1.41/8.02
StableCodec	1	2.54/105.18	6.19/200.03	1.07/6.70
CoD	1	1.65/18.12	6.75/62.19	0.66/4.21
DiT-IC	1	2.76/97.62	4.54/151.89	1.07/5.07
Gen2-IC variants				
Gen2-IC-SD	1	2.32/27.87	6.39/103.05	1.00/2.70
Gen2-IC-SANA	1	2.51/31.75	5.54/102.75	1.98/8.19
Gen2-IC-FLUX	1	2.33/21.82	38.83/193.58	12.17/32.75
Gen2-IC-Qwen	1	3.59/24.05	59.96/355.11	20.64/56.40

Table 4: Computational complexity on Kodak at 512×768, measured on a single NVIDIA A100 GPU. S: scheduler updates; C/T: MMAC/pixel and ms/image; P/M: parameters (B) and peak memory (GiB).

t_{inj}	$\beta^\star$	w/o LoRA	w/ LoRA
1	0.01	0.1941	0.1847
5	0.01	0.1919	0.1812
10	0.05	0.1890	0.1765
15	0.05	0.1858	0.1748
20	0.05	0.1827	0.1737
25	0.05	0.1793	0.1720
30	0.15	0.1768	0.1711
35	0.25	0.1743	0.1696
40	0.45	0.1728	0.1668
45	0.80	0.1722	0.1577
50	0.99	0.1716	**0.1427**

Table 5: Near-terminal injection ablation. DISTS (↓) for Gen2-IC-Qwen at $\mathrm{qp} = 0$ on Kodak, using $\beta^\star$ selected on the validation set for each t_{inj}. We compare the frozen backbone with its LoRA-adapted counterpart after 100 updates; $t_{\mathrm{inj}} = 50$ is the native endpoint of the generation schedule.

Later injection monotonically improves both variants, from $0.1941/0.1847$ at $t_{\mathrm{inj}} = 1$ to $0.1716/0.1427$ at the native endpoint $t_{\mathrm{inj}} = 50$. The consistent endpoint optimum before and after adaptation supports near-terminal correction, with LoRA further amplifying its benefit.

5 Conclusion

We introduced Gen2-IC, a simple, extensible interface coupling compression and generation through separately optimized Latent Compression and one-step Latent Transport. Using decoded latents as near-terminal states, its direct, single-path design supports fast encoding and multiple rates without extra conditioning or task-specific backbone modifications. The shared formulation applies across SD-2.1, SANA-1.5, FLUX.1-dev, and Qwen-Image-2512: the matched SD-2.1 comparison isolates interface gains from model scale, while FLUX and Qwen demonstrate how stronger priors further advance ultra-low-rate compression.

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The main paper presents the core idea and performance of Gen2-IC. In this supplementary material, we provide additional experimental details, results, and analyses, including:

- **Section A: Detailed Method.** Inference, native scheduler conversion, codec architecture, and transport objective.
- **Section B: Implementation and Reproducibility.** Training, validation, evaluation, and baseline settings.
- **Section C: Additional Results.** Matched prior contributions, complete quality curves, human preference, and multi-resolution efficiency.
- **Section D: Additional Ablations.** LoRA activation/rank and conditioning without prompt transmission.
- **Section E: Visual Results.** Matched-rate source-fidelity comparisons and module-level evidence in this appendix.

A Detailed Method

A.1 End-to-End Inference

Algorithm S1 Gen2-IC inference at operating point q

Require: Image x; operating point q; Gen2-IC modules; native scheduler $\mathcal{S}$; fixed $\tau_\star$ and β_q
Ensure: Bitstream $\mathcal{B}$ and reconstruction $\hat{x}_1$

$l \leftarrow \mathcal{E}_e(x)$
$y \leftarrow g_a(l, q_e^{(q)})$
$\mathcal{B} \leftarrow \mathrm{EntropyEncode}(y; \phi)$
$\hat{y} \leftarrow \mathrm{EntropyDecode}(\mathcal{B}; \phi)$
$\hat{l}_0 \leftarrow g_s(\hat{y}, q_d^{(q)})$
$(s_\star, m_\star) \leftarrow \mathrm{ConstructState}_{\mathcal{S}}(\hat{l}_0, \tau_\star)$
$p_\star \leftarrow P_\theta(s_\star, \tau_\star; m_\star)$
$s_1 \leftarrow \mathrm{Step}_{\mathcal{S}}(p_\star, \tau_\star, s_\star)$
$\tilde{l}_1 \leftarrow \mathrm{ToVAELatent}_{\mathcal{S}}(s_1, m_\star)$
$\hat{l}_1 \leftarrow \hat{l}_0 + \beta_q(\tilde{l}_1 - \hat{l}_0)$
$\hat{x}_1 \leftarrow \mathcal{E}_d(\hat{l}_1)$
return $\mathcal{B}, \hat{x}_1$

Algorithm S1 expands the Δ_θ interface in the main paper. The encoder comprises the frozen VAE encoder and Latent Compression. The decoder reconstructs $\hat{l}_0$, converts it to the last nontrivial native state, and applies one scheduler update. Its output $\tilde{l}_1$ defines $\Delta_\theta = \tilde{l}_1 - \hat{l}_0$, scaled by β_q. Thus, "one-step" denotes exactly one native scheduler update; end-to-end latency is reported in the main paper and Table S9.

The metadata $m_\star$ stores layout, image IDs, sequence shape, prompt masks, and schedule indices; deterministic seeds generate construction noise. The positive prompt is empty, leaving $\hat{l}_0$ as the image-specific signal without an image prompt, semantic feature, or auxiliary bitstream.

A.2 Native Scheduler Conversion

Table S7 details SD-2.1 (Rombach et al. 2022), SANA-1.5 (Xie et al. 2025), FLUX.1-dev (Black Forest Labs 2024), and Qwen-Image-2512 (Wu et al. 2025). Each uses its last nontrivial native interval while retaining model-specific state, prediction, scheduler, adapter, and latent operations.

Latent coordinates. SD uses $l = z\, s_{\mathrm{VAE}}$; SANA applies analogous DC-AE scaling. FLUX uses $l = (z - s_{\mathrm{shift}})s_{\mathrm{scale}}$, and Qwen uses $l_c = (z_c - \mu_c)/\sigma_c$. Both stages share these native coordinates. SD and SANA retain BCHW layout; FLUX and Qwen apply $B \times 16 \times H_l \times W_l \to B \times (H_l W_l/4) \times 64$ packing before prediction and invert it afterward; Qwen also retains its singleton temporal dimension.

Terminal-state construction. We use native 20-step SD/SANA and 50-step FLUX/Qwen schedules, executing only their final update. FLUX/Qwen retain the sequence-length shift and use sigmas from 1 to $1/50$ with `scale_noise`. All variants use seed 42, the official default negative prompt, and an empty positive prompt; CFG is 5 for Qwen and 1 otherwise. Outputs return to VAE coordinates before $\hat{l}_1 = (1 - \beta_q)\hat{l}_0 + \beta_q \tilde{l}_1$.

A.3 Latent Codec Architecture

Trainable module. The native VAE and backbone remain unchanged. Latent Compression contains analysis/synthesis transforms g_a, g_s and entropy model ϕ, using depthwise-convolution blocks for parameter-efficient nonlinear capacity. Its representation y is split into four quadtree groups whose Gaussian parameters (μ_i, σ_i) are predicted from decoded groups. Channel-wise $q_e^{(q)}, q_d^{(q)}$ and entropy-model quantization scales provide five rates in one codec.

A.4 What the Transport Objective Learns

After freezing Latent Compression, we back-propagate $\mathcal{L}_1 + \mathcal{L}_{\mathrm{per}} + \mathcal{L}_{\mathrm{adv}}$ through the frozen VAE decoder and near-terminal update to LoRA. This source-referenced loss directly trains a residual correction at the decoded latent. With the VAE and codec fixed, transport improves PSNR, MS-SSIM, LPIPS, and DISTS. Source anchoring and terminal placement jointly constrain refinement to remain faithful while recovering compression-impaired details.

B Implementation and Reproducibility

B.1 Training and Validation Protocol

We retain the 10,000 highest-resolution images from each of ImageNet (Deng et al. 2009) and OpenImages (Kuznetsova et al. 2020). The 30 lowest-resolution images in this combined pool form validation; all others form training. Latent Compression samples random 256×256 crops and $q \in \{0, \ldots, 4\}$, with rate weight log-linearly interpolated over $[0.1, 1.0]$; Latent Transport uses 512×512 crops. We set the Python, NumPy, and PyTorch training seeds to 42 for every variant. Transport freezes the codec, VAE, and backbone and optimizes attention LoRA only. Each adapter (Hu et al. 2022) uses rank 16, $\alpha = 32$, zero dropout, and targets `to_q`, `to_k`, `to_v`, and `to_out.0`.

Each variant trains one adapter at $\mathrm{qp} = 0$ and reuses it across rates. Validation selects its five β_{rate} values within the codec's bitrate regime. Table S1 reports the variant-specific β_{rate} values, frozen before all four test sets without per-image or per-dataset adjustment.

The stages are optimized sequentially: Latent Compression is frozen before LoRA adaptation, separating coding from generative refinement without joint training.

Variant	β_{rate} for $\mathrm{qp} = 0, 1, 2, 3, 4$
Gen2-IC-SD	$\{0.95, 0.80, 0.60, 0.45, 0.30\}$
Gen2-IC-SANA	$\{0.95, 0.80, 0.60, 0.45, 0.30\}$
Gen2-IC-FLUX	$\{0.99, 0.95, 0.65, 0.55, 0.40\}$
Gen2-IC-Qwen	$\{0.99, 0.85, 0.55, 0.45, 0.30\}$

Table S1: Validation-selected β_{rate} values, fixed before testing.

B.2 Evaluation Datasets and Metrics

Following prior codecs, we use these standard benchmarks to align evaluation sets and enable direct baseline comparison.

(a) Test datasets

Dataset	Images	Resolution
Kodak (Kodak 1993)	24	768×512
Tecnick (Asuni and Giachetti 2014)	100	1200×1200
DIV2K val. (Agustsson and Timofte 2017)	100	$\sim$2K
CLIC 2020 Pro. (Toderici et al. 2020)	250	up to 2K

(b) Reported metrics

Category	Metric	Better
Rate	bpp	$\downarrow$
Fidelity	PSNR; MS-SSIM	$\uparrow$
Perceptual	LPIPS-VGG; DISTS	$\downarrow$
Distributional	FID; KID	$\downarrow$
No-reference	MUSIQ; CLIP-IQA	$\uparrow$
Text consistency	CER; WER	$\downarrow$
Face consistency	face-ROI cosine similarity	$\uparrow$
Curve summary	BD-rate	$\downarrow$

Table S2: Evaluation datasets and metrics; FID/KID follow GLC and omit Kodak.

Each $(\text{variant}, \text{dataset}, \mathrm{qp})$ uses one deterministic run with seed 42. RGB inputs are replicate-padded to multiples of 32 (SD/FLUX/Qwen) or 128 (SANA), coded, and cropped before scoring.

Rate and image metrics. Rate is $8N_{\mathrm{IP}}/(HW)$, where N_{IP} is the emitted intra-picture stream size; the reusable sequence header is excluded. PSNR is $-10\log_{10}\mathrm{MSE}$ on RGB $[0, 1]$ tensors. Saved images are scored by `torchmetrics` MS-SSIM (Wang, Simoncelli, and Bovik 2003) (`data_range=1`) and LPIPS-VGG (Zhang et al. 2018) with $[0, 1] \to [-1, 1]$ normalization. DISTS (Ding et al. 2020), MUSIQ (Ke et al. 2021), and CLIP-IQA (Wang, Chan, and Loy 2023) use `pyiqa`. The latter two score reconstructions only; all per-image scores are averaged by dataset and operating point.

Distributional metrics. FID/KID (Heusel et al. 2017; Bińkowski et al. 2018) follow GLC (Jia et al. 2024). We extract non-overlapping 256×256 patches on regular and $(128, 128)$-shifted grids, quantize to uint8, and update `torchmetrics` Inception-2048 statistics. KID uses 100 subsets of size $\min(50, N_{\mathrm{patch}})$. Each dataset/rate is scored over its full patch distribution. FID/KID use Tecnick, DIV2K, and CLIC 2020, whose sizes support patch-distribution estimates; Kodak contains only 24 images and is not used for these metrics.

BD-rate. BD-rate (Bjontegaard 2001) uses PCHIP interpolation of log-rate over each pair's common metric interval, without extrapolation. Samples are metric-sorted; duplicate metric values are merged by mean rate, and all remaining overlapping samples are retained. Comparisons require a nonempty common interval. Negative values denote bitrate savings at matched quality.

Recognizer-based ROI construction. EasyOCR uses CRAFT (Baek et al. 2019) with Chinese/English recognition. Source text boxes require confidence ≥ 0.3 and at least two alphanumeric/CJK characters. InsightFace `buffalo_l` uses SCRFD (Guo et al. 2022) at 640×640 and threshold 0.5. Detectors run once on sources, and fixed boxes are reused across reconstructions. Table S3 reports 348 text boxes in 86 images and 215 face boxes in 74 images; 56 and 24 images contain multiple boxes, with per-image maxima of 27 and 54.

(a) Text ROIs

Dataset	Images	ROIs	Multi	Max/image
Kodak	2	5	2	3
Tecnick	10	24	6	5
DIV2K	16	64	13	11
CLIC 2020	58	255	35	27
Total	86	348	56	27

(b) Face ROIs

Dataset	Images	ROIs	Multi	Max/image
Kodak	7	16	3	8
Tecnick	1	1	0	1
DIV2K	17	57	7	19
CLIC 2020	49	141	14	54
Total	74	215	24	54

Table S3: Source ROIs. Multi counts multi-region images; Max/image is the largest per-image count.

Text consistency. Each source text box is expanded by 5% and mapped to the reconstruction. EasyOCR/CRNN (Shi, Bai, and Yao 2017) outputs with confidence ≥ 0.3 are ordered by reading order, lowercased, and restricted to alphanumeric/CJK characters. CER removes spaces; WER tokenizes them. Levenshtein distance is source-length normalized. Scores are averaged first by image, then by dataset.

Face consistency. Source face boxes and five-point landmarks are mapped to reconstructions. ArcFace (Deng et al. 2019) embeds aligned 112×112 crops; when direct alignment is unavailable, detection is repeated within a 20%-

enlarged fixed box. Cosine similarity of ℓ_2-normalized embeddings is averaged by image, then dataset. The evaluation includes boxes down to 11×16 pixels; minimum/median SCRFD scores are 0.501/0.783. CER/WER use source EasyOCR outputs, while face consistency uses source/reconstruction ArcFace embeddings within the fixed ROIs.

B.3 Baseline and Complexity Settings

Classical codec. VTM-23.10 (VTM-23.10 2025) uses one-frame all-intra coding with 8-bit YUV 4:2:0 input and the official configuration in Table S4.

Setting	Value
Executable	`EncoderApp`
Configuration	`encoder_intra_vtm.cfg`
Input / reconstruction	`input.yuv` / `output.yuv`
Bitstream	`output.bin`
Resolution	`-wdt [width] -hgt [height]`
Quality point	`-q [QP]`
Bit depth / chroma	8-bit / YUV 4:2:0
Frames / frame rate	1 / 1

Table S4: VTM-23.10 all-intra evaluation configuration.

Neural codecs and complexity. DCVC-RT also runs in intra mode; neural-codec rates use actual entropy-coded streams. `calflops` measures MACs with 1 FLOP $=$ 2 MACs. A100 latency includes arithmetic coding and averages ten runs after four warm-ups. We disable VAE tiling/slicing, sharding, sparse attention, and deployment-specific backbone optimization.

C Additional Quantitative Results

C.1 Matched Generative-Prior Contribution

Controlled comparison. Table S5 extends the main paper's matched FLUX/Qwen ablation to SD/SANA. Each pair fixes the native VAE and codec, changing only Latent Transport, which improves every metric. Across four native latent spaces and formulations, these matched comparisons establish consistent within-variant gains from Latent Transport.

	BD-rate (%) ↓			
Variant	PSNR	MS-SSIM	LPIPS	DISTS
SD: Latent Compression	114.52	39.50	-48.88	-65.14
SD: + Latent Transport	71.82	25.53	-62.65	-72.64
SANA: Latent Compression	144.77	54.94	-59.91	-73.36
SANA: + Latent Transport	114.28	39.89	-69.72	-79.58

Table S5: Matched SD/SANA Kodak ablation relative to MS-ILLM, using the latent-codec and corresponding one-step LoRA curves with each VAE and codec fixed.

Interpretation. These matched gains directly isolate Latent Transport and show consistent distortion and perceptual improvements across all four native latent spaces.

C.2 Additional Quality Metrics

Complementary curves. Figures S3 and S4 provide complete PSNR, MS-SSIM, LPIPS, DISTS, MUSIQ (Ke et al. 2021), and CLIP-IQA (Wang, Chan, and Loy 2023) curves on all four benchmarks, together with FID/KID on Tecnick, DIV2K, and CLIC 2020. They extend the main results across distortion, perceptual, no-reference, and distributional criteria.

C.3 User Study

Protocol. Following StableCodec's rate-matched protocol (Zhang et al. 2025), 33 participants select the best reconstruction among randomly ordered methods for each of 24 Kodak images, yielding $33 \times 24 = 792$ top-1 votes.

Result. Gen2-IC-Qwen ranks first with 268 votes (33.84%), 1.65× the runner-up and 3.57× Gen2-IC-Base (pixel codec). This preference gap complements the quantitative results and corroborates Gen2-IC's human-perceived reconstruction quality.

Method	Top-1 votes	Share (%)
Gen2-IC-Qwen	**268**	**33.84**
OneDC	162	20.45
StableCodec	100	12.63
Gen2-IC-Base (pixel codec)	75	9.47
GLC	71	8.96
OSCAR	39	4.92
MS-ILLM	34	4.30
DiffEIC (Li et al. 2025a)	27	3.41
ResULIC	16	2.02

Table S6: Blind top-1 preference on 24 rate-matched Kodak cases; shares sum to 100%.

C.4 Runtime and Memory Across Resolutions

Memory scaling. Table S8 reports total peak-reserved and stage-incremental allocated memory. Its 512×768 maxima match the peak memory in main Table 4. At 1440×2560, Latent Transport adds only 0.23–1.59 GiB beyond resident allocation across all variants, despite $9.4\times$ more pixels. Thus, transport has modest resolution-dependent memory overhead even for large priors.

Runtime scaling. Table S9 reports end-to-end runtime. At both 1080×1920 and 1440×2560, every Gen2-IC variant encodes faster than all baselines with reported runtimes, and Gen2-IC-SANA decodes fastest. At the largest resolution, matched Gen2-IC-SD is $10.7\times$ faster in encoding and $4.6\times$ faster in decoding than OSCAR, confirming that the interface gain scales beyond Kodak resolution.

D Additional Ablations

D.1 LoRA Activation and Rank

Prior activation versus rank. Table S10 separates prior invocation from adaptation in Gen2-IC-Qwen. LoRA produces substantial gains over both VAE-only and frozen prior

correction. Rank 16, $\alpha = 32$ performs best overall, while ranks 8 and 32 retain most gains. For Gen2-IC-Qwen, this confirms that lightweight prior activation matters more than the exact rank.

D.2 Prompt Conditioning

Protocol and overhead. We compare empty, generic, and compression-oriented prompts. Qwen3-VL-30B-A3B-Instruct (Bai et al. 2025) generates the latter two at temperature 0 ($\leq$ 4096 tokens); lossless `.xz` overhead averages 0.0002/0.002 bpp for generic/compression prompts.

Result. For Gen2-IC-Qwen, the empty prompt achieves the strongest overall results in Table S11 without prompt transmission. This confirms that the decoded latent provides sufficient image-specific guidance and supports Gen2-IC without extra transmitted conditions. Prompt Boxes D.1 and D.2 give the exact templates. Examples are included at original resolution in Figures S5 and S6.

Prompt Box D.1: Generic Prompt Template

Role. You are a prompt-writer for image reconstruction.
Goal. Produce a compact instruction that preserves all visually certain details needed to reconstruct the same content, style, and composition.
Requirements.

- Cover certain subjects, layout, background, colors, lighting, materials, viewpoint, and style.
- Prefer factual nouns and attributes over poetic adjectives.
- State an obvious style (e.g., photo, 3D, anime, flat icon, or line art) in 1–3 tokens and avoid artist lists.

Prompt Box D.2: Compression Prompt Template

Role. You are a prompt-writer for image-to-image refinement.
Input. Base image B is the coarse reconstruction and target image A is the source.
Goal. Describe only the changes needed to transform B into A while preserving matched content and composition.
Requirements.

- Keep the prompt compact and explicitly preserve framing/composition.
- Do not re-describe the full image.
- Focus on missing objects, corrected colors, textures, lighting, fine details, sharpness, and artifact removal.

These ablations use Gen2-IC-Qwen. Together, the ablations validate lightweight LoRA activation and operation without transmitted prompts.

E Visual Results

Figures S7–S13 compare matched-rate reconstructions on face-, structure-, and text-sensitive examples, with actual bpp and magnified regions. Figures S14– S16 isolate Gen2-IC-Base (pixel codec), VAE reconstruction, frozen-prior correction, and LoRA-adapted Latent Transport, corroborating the matched quantitative ablations. The original-resolution panels are embedded in this appendix.

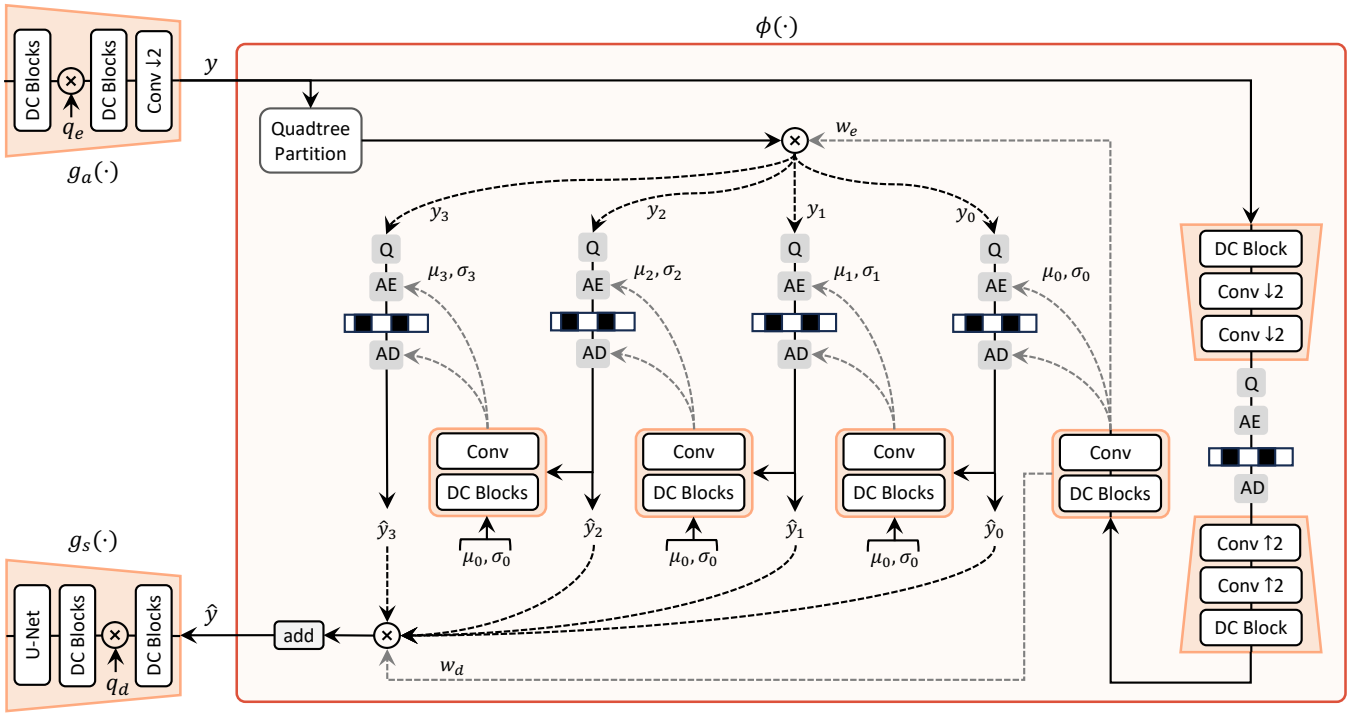


Figure S1: Latent Compression: analysis/synthesis transforms and a four-group quadtree entropy model with arithmetic encoding (AE) and decoding (AD).

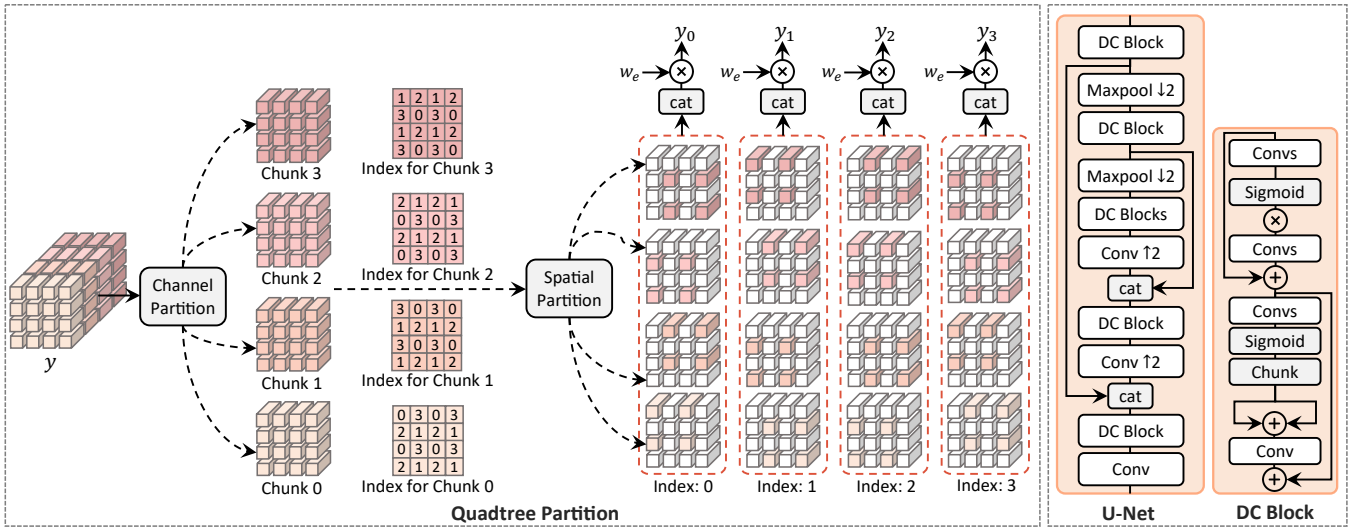


Figure S2: Detailed quadtree partition, U-Net transform, and depthwise convolution (DC) blocks used by Latent Compression.

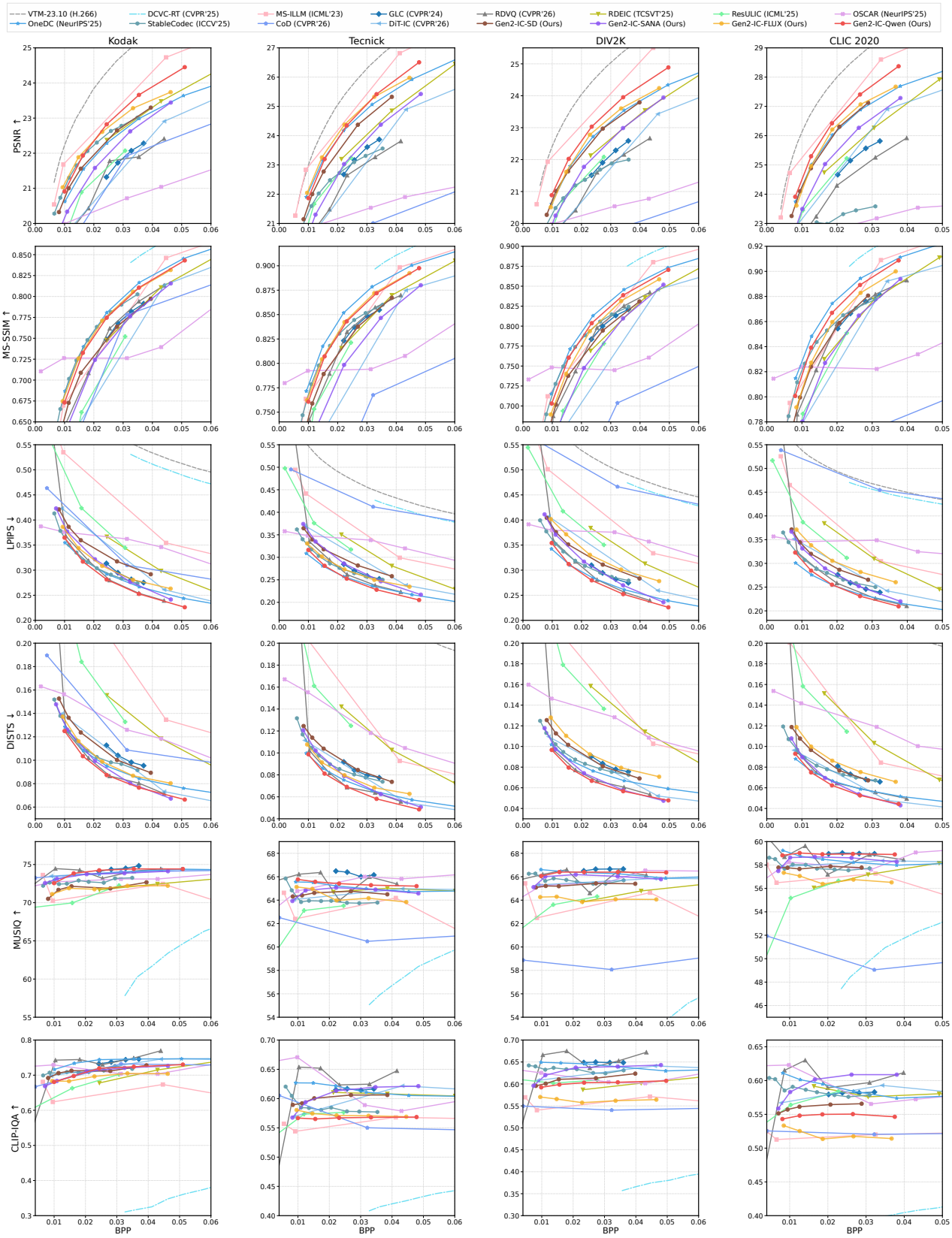


Figure S3: PSNR, MS-SSIM, LPIPS, DISTS, MUSIQ, and CLIP-IQA rate–quality curves on all four benchmarks.

Variant	VAE/codec state	State construction and native update
Gen2-IC-SD	Scaled BCHW latent, $4 \times H/8 \times W/8$	Add native noise at the penultimate state; scale the model input; predict with the SD-2.1 U-Net; take the final scheduler step
Gen2-IC-SANA	Scaled BCHW latent, $32 \times H/32 \times W/32$	Construct the terminal noisy state using native noise/sigma conversion; predict with the SANA Transformer; take the native step
Gen2-IC-FLUX	Shifted/scaled 16-channel latent; 2×2 pack to 64-D tokens	Use the resolution-shifted flow schedule, image IDs, and empty-prompt embeddings; take one native step; unpack to BCHW
Gen2-IC-Qwen	Channel-standardized 16-channel latent; 2×2 pack to 64-D tokens	Use the resolution-shifted flow schedule, image shapes, and prompt masks; apply true CFG $= 5$; take one native step; unpack to BCHW

Table S7: Model-specific Latent Transport interfaces; each executes one final native scheduler update.

Variant	Stage	512×768	1080×1920	1440×2560
Gen2-IC-SD	VAE encoding	1.02 / 0.56	4.56 / 3.00	7.79 / 5.29
	Latent codec encoding	0.49 / 0.07	1.07 / 0.38	1.69 / 0.67
	Latent codec decoding	0.46 / 0.06	0.86 / 0.34	1.28 / 0.59
	VAE decoding	1.45 / 0.89	5.97 / 4.72	10.32 / 8.33
	Latent Transport	2.70 / 0.10	3.18 / 0.51	3.75 / 0.90
Gen2-IC-SANA	VAE encoding	1.53 / 0.66	5.05 / 3.70	8.38 / 6.58
	Latent codec encoding	0.78 / 0.01	0.78 / 0.05	0.88 / 0.09
	Latent codec decoding	0.78 / 0.01	0.78 / 0.05	0.87 / 0.08
	VAE decoding	1.58 / 0.61	5.54 / 3.42	9.29 / 6.08
	Latent Transport	8.19 / 0.03	8.38 / 0.13	8.53 / 0.23
Gen2-IC-FLUX	VAE encoding	1.13 / 0.56	4.64 / 3.00	8.38 / 5.29
	Latent codec encoding	0.40 / 0.04	0.66 / 0.19	0.92 / 0.34
	Latent codec decoding	0.38 / 0.03	0.56 / 0.17	0.90 / 0.30
	VAE decoding	1.41 / 0.89	5.89 / 4.73	10.63 / 8.35
	Latent Transport	32.75 / 0.21	38.77 / 0.91	40.14 / 1.56
Gen2-IC-Qwen	VAE encoding	1.51 / 0.99	7.47 / 5.26	12.08 / 9.28
	Latent codec encoding	0.44 / 0.04	0.72 / 0.19	1.04 / 0.34
	Latent codec decoding	0.43 / 0.03	0.62 / 0.17	0.84 / 0.30
	VAE decoding	2.21 / 1.55	9.86 / 8.23	17.04 / 14.52
	Latent Transport	56.40 / 0.17	56.64 / 0.90	56.98 / 1.59

Table S8: GPU memory across resolutions (GiB); entries report total peak reserved / stage-incremental allocated memory.

Method	512×768		1080×1920		1440×2560	
	Enc.	Dec.	Enc.	Dec.	Enc.	Dec.
VAE-based methods						
RDVQ (Jiang et al. 2026)	408.24	412.44	704.64	809.99	953.80	1130.17
Diffusion-based methods						
DiffEIC (Li et al. 2025a)	210.18	4661.74	–	>10 s	–	>15 s
RDEIC (Li et al. 2025b)	157.25	426.68	683.20	2296.58	1436.00	5963.77
ResULIC (Ke et al. 2025)	96.24	428.32	481.75	3702.45	985.24	10028.48
OSCAR (Guo et al. 2025)	65.13	163.77	857.68	1109.37	2982.82	2751.47
OneDC (Xue et al. 2025b)	100.50	235.03	366.32	763.99	680.47	1452.33
StableCodec (Zhang et al. 2025)	105.18	200.03	641.76	1657.43	1085.39	3161.54
CoD (Jia et al. 2026)	18.12	62.19	166.72	439.11	635.46	1866.48
DiT-IC (Shi et al. 2026)	97.62	151.89	480.22	750.38	718.17	1111.59
Gen2-IC variants						
Gen2-IC-SD	27.87	103.05	144.79	323.19	279.83	600.14
Gen2-IC-SANA	31.75	102.75	132.58	234.72	213.82	373.40
Gen2-IC-FLUX	21.82	193.58	133.80	923.57	264.12	1807.57
Gen2-IC-Qwen	24.05	355.11	130.99	2189.06	247.65	4956.31

Table S9: Runtime (ms/image). Enc./Dec. cover the full coding/reconstruction pipelines under the same A100 protocol.

Qwen variant	PSNR BD-rate ↓				LPIPS BD-rate ↓				DISTS BD-rate ↓			
	Kod.	Tec.	DIV2K	CLIC	Kod.	Tec.	DIV2K	CLIC	Kod.	Tec.	DIV2K	CLIC
VAE only	95.35	95.11	120.19	92.67	-56.57	-44.31	-52.16	-49.12	-65.68	-49.34	-62.61	-47.73
Transport, frozen prior	87.88	87.40	111.71	84.54	-57.42	-46.04	-53.37	-50.69	-65.16	-49.83	-62.72	-48.13
LoRA $r = 8, \alpha = 16$	48.77	49.91	73.98	53.64	-74.00	-65.91	-68.94	-66.22	-79.51	-65.94	-75.18	-65.77
LoRA $r = 16, \alpha = 32$	44.66	44.23	67.41	46.92	-75.84	-69.25	-72.21	-70.48	-81.21	-71.18	-79.00	-70.66
LoRA $r = 32, \alpha = 64$	45.89	46.86	70.04	50.08	-75.03	-66.55	-68.58	-66.62	-80.32	-66.10	-73.66	-64.99

Table S10: Gen2-IC-Qwen LoRA activation/rank ablation. BD-rate (%) is relative to MS-ILLM.

Setting	PSNR BD-rate (%) ↓				LPIPS BD-rate (%) ↓				DISTS BD-rate (%) ↓			
	Kod.	Tec.	DIV2K	CLIC	Kod.	Tec.	DIV2K	CLIC	Kod.	Tec.	DIV2K	CLIC
No transport	0	0	0	0	0	0	0	0	0	0	0	0
Frozen + Empty	-3.89	-3.92	-3.93	-3.92	-1.69	-2.57	-2.05	-2.51	1.06	-0.97	-0.35	-0.79
Frozen + Generic	7.43	0.90	0.10	0.49	8.28	1.82	0.58	1.42	8.15	1.80	0.98	1.40
Frozen + Compression	9.66	1.06	0.08	0.49	10.78	2.00	0.56	1.44	10.89	2.01	1.05	1.54
LoRA + Empty	-25.37	-24.44	-22.02	-21.04	-41.89	-43.11	-40.08	-37.73	-43.13	-42.22	-43.49	-40.13
LoRA + Generic	-14.19	-22.15	-20.97	-19.21	-23.75	-40.29	-39.83	-35.39	-19.77	-40.01	-45.95	-38.92
LoRA + Compression	-8.82	-19.77	-20.09	-18.49	-13.87	-35.12	-36.68	-32.48	-7.89	-29.62	-38.15	-32.75

Table S11: Gen2-IC-Qwen prompt/LoRA ablation. BD-rate (%) is relative to no transport and includes prompt bits.

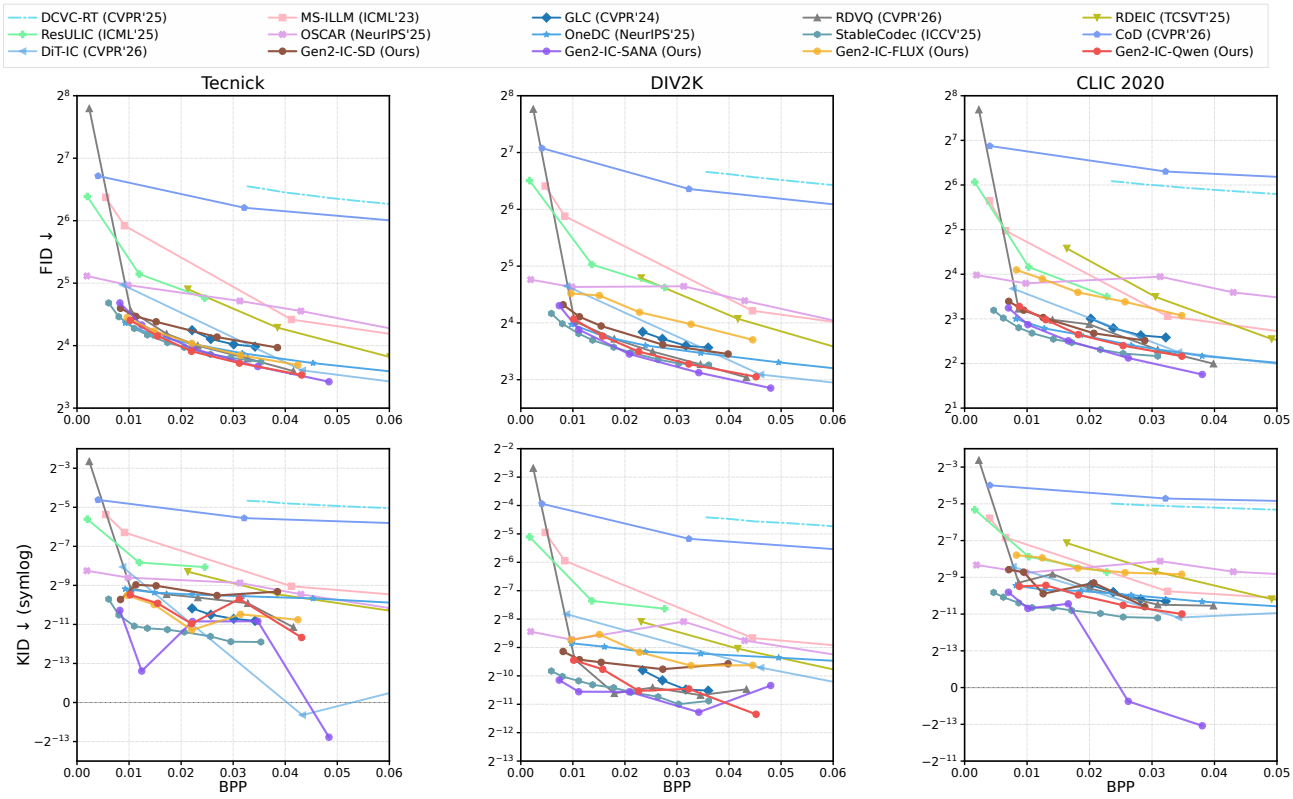
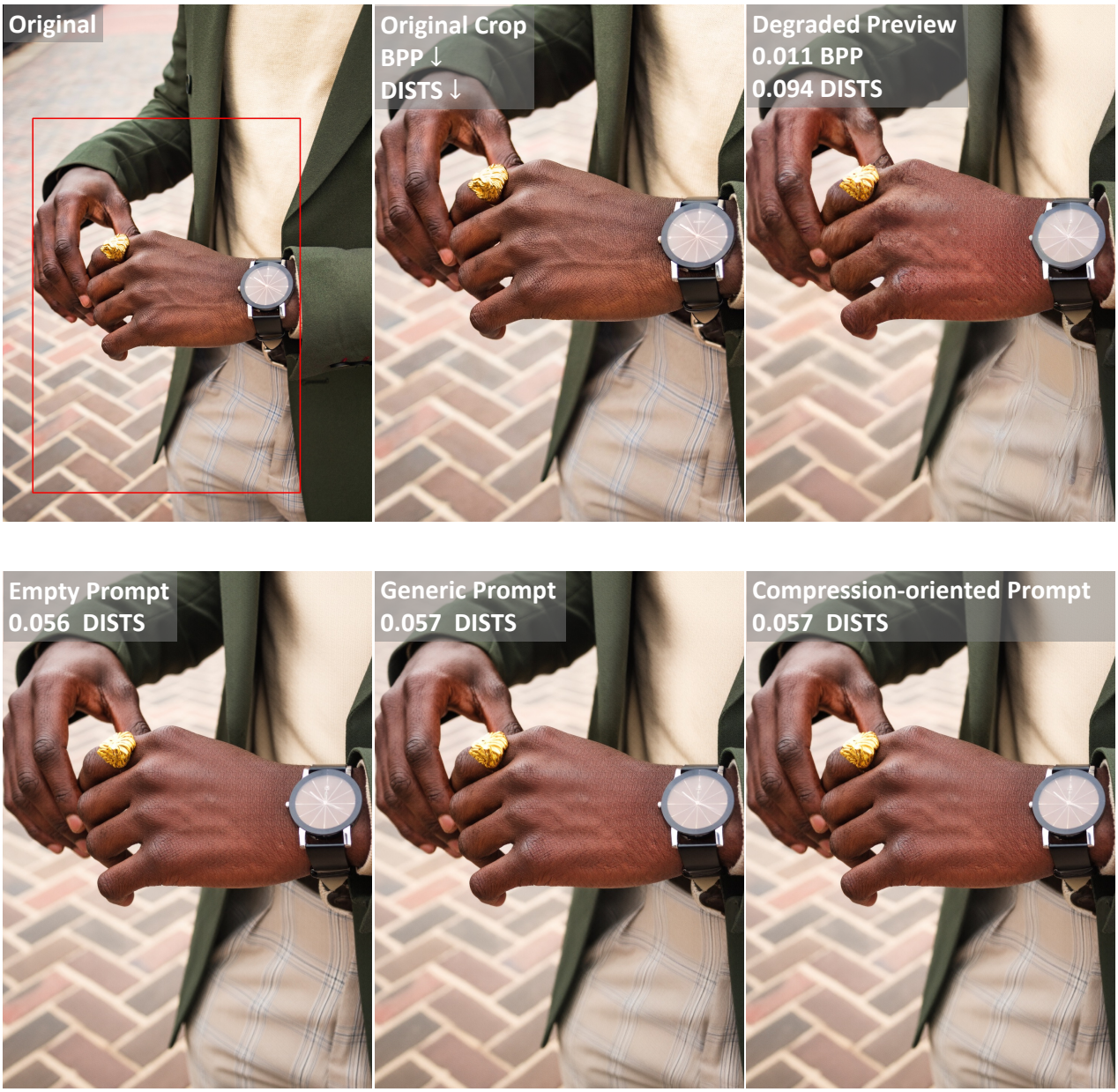


Figure S4: FID/KID rate–quality curves on Tecnick, DIV2K, and CLIC 2020.



Generic Prompt: *"close-up of a man's hands, wearing a green blazer, beige sweater, and plaid pants, adjusting a gold lion ring on his finger, with a black watch on his wrist, standing on a brick sidewalk"*

Compression-oriented Prompt: *"replace the blurry, distorted hands and watch with sharp, realistic hands holding a gold lion ring, add plaid pants, and refine the background to a clear brick pavement while keeping the same framing and composition"*

Figure S5: Prompt-conditioned comparisons; the empty prompt preserves source content without transmitted prompt bits.

Generic Prompt: *"a young woman with long brown hair, beige beret, round glasses, brown coat, white polka-dot dress, sitting on plaid blanket outdoors, chin on hand, looking at camera, soft natural light, shallow depth of field, autumn background "*

Compression-oriented Prompt: *"refine background to show grass and fallen leaves, add plaid blanket, improve lighting and sharpness, enhance skin texture and eye detail, keep same framing and composition"*

Figure S6: Prompt-conditioned comparisons; decoded-latent guidance remains effective without transmitted prompt bits.

Figure S7: Matched-rate comparison on a face-containing scene. The inset highlights profile and hair details.

Figure S8: Matched-rate comparison on a portrait scene. The inset highlights facial details.

Figure S9: Matched-rate comparison on a multi-person scene. The inset highlights facial reconstruction.

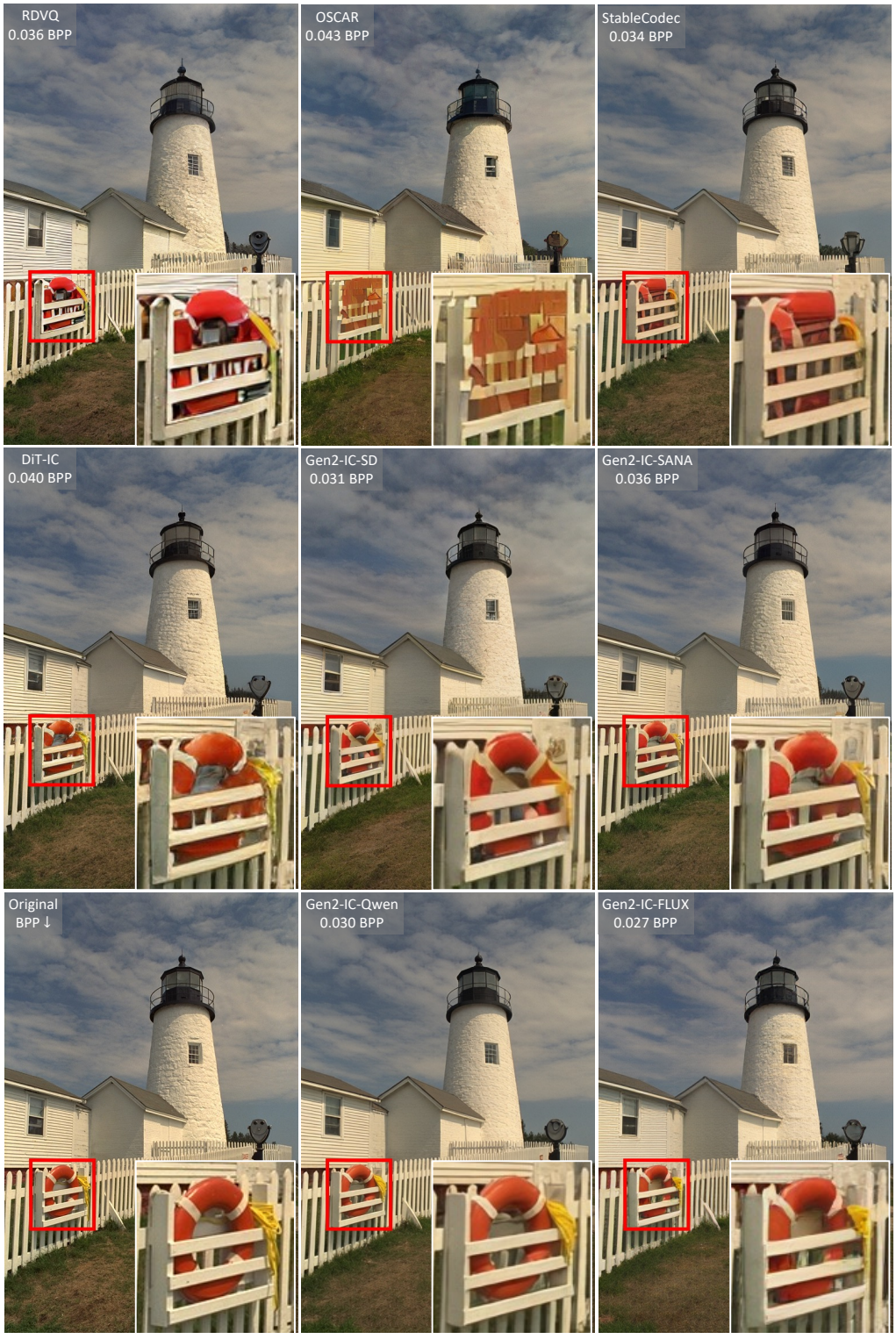


Figure S10: Matched-rate comparison on a structured scene. The inset highlights the small foreground object.

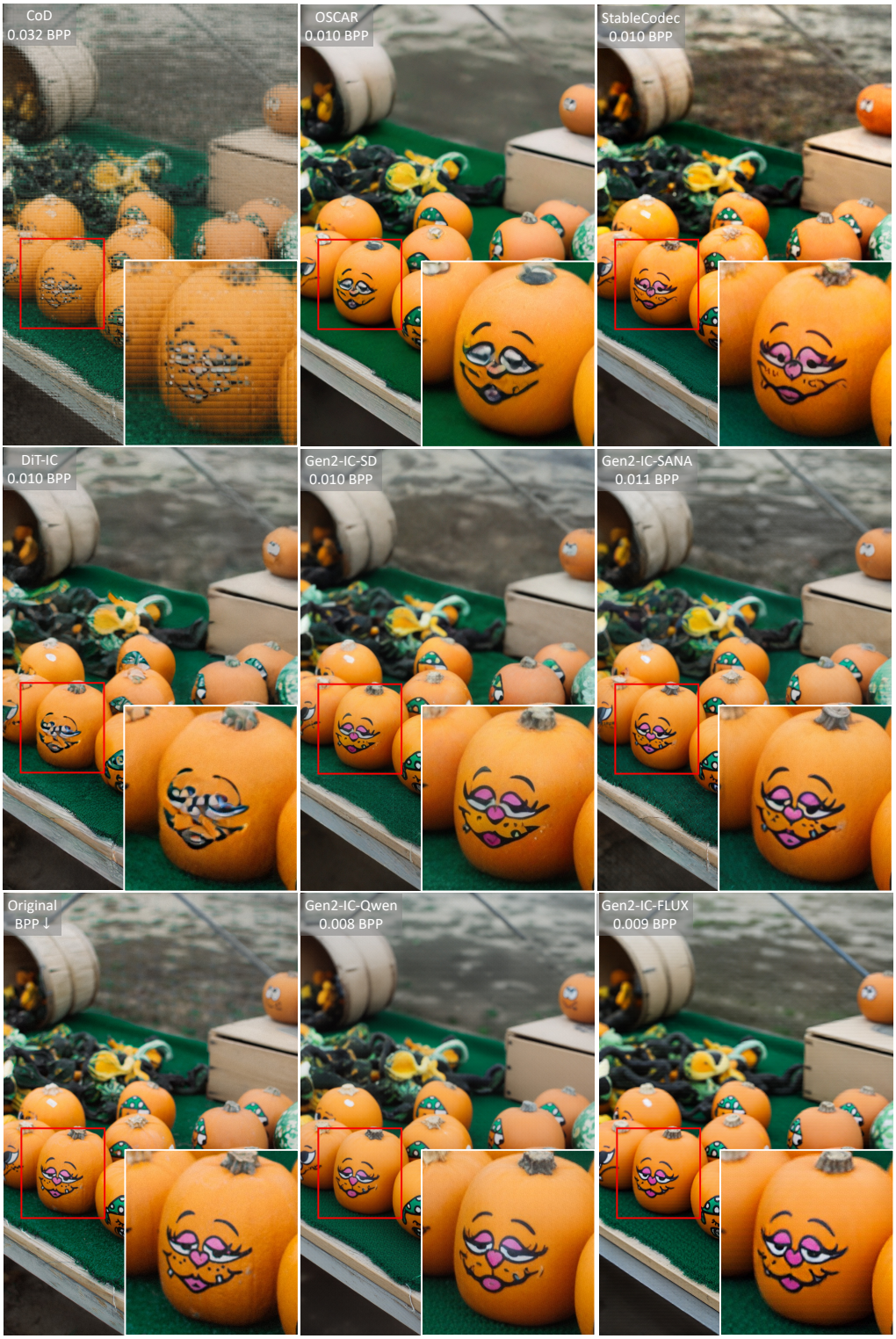


Figure S11: Matched-rate comparison on a detail-rich scene. The inset highlights the painted pattern.

Figure S12: Matched-rate comparison on a text- and logo-rich scene. The inset highlights fine line structures.

Figure S13: Matched-rate comparison on a product scene. The inset highlights small printed text.

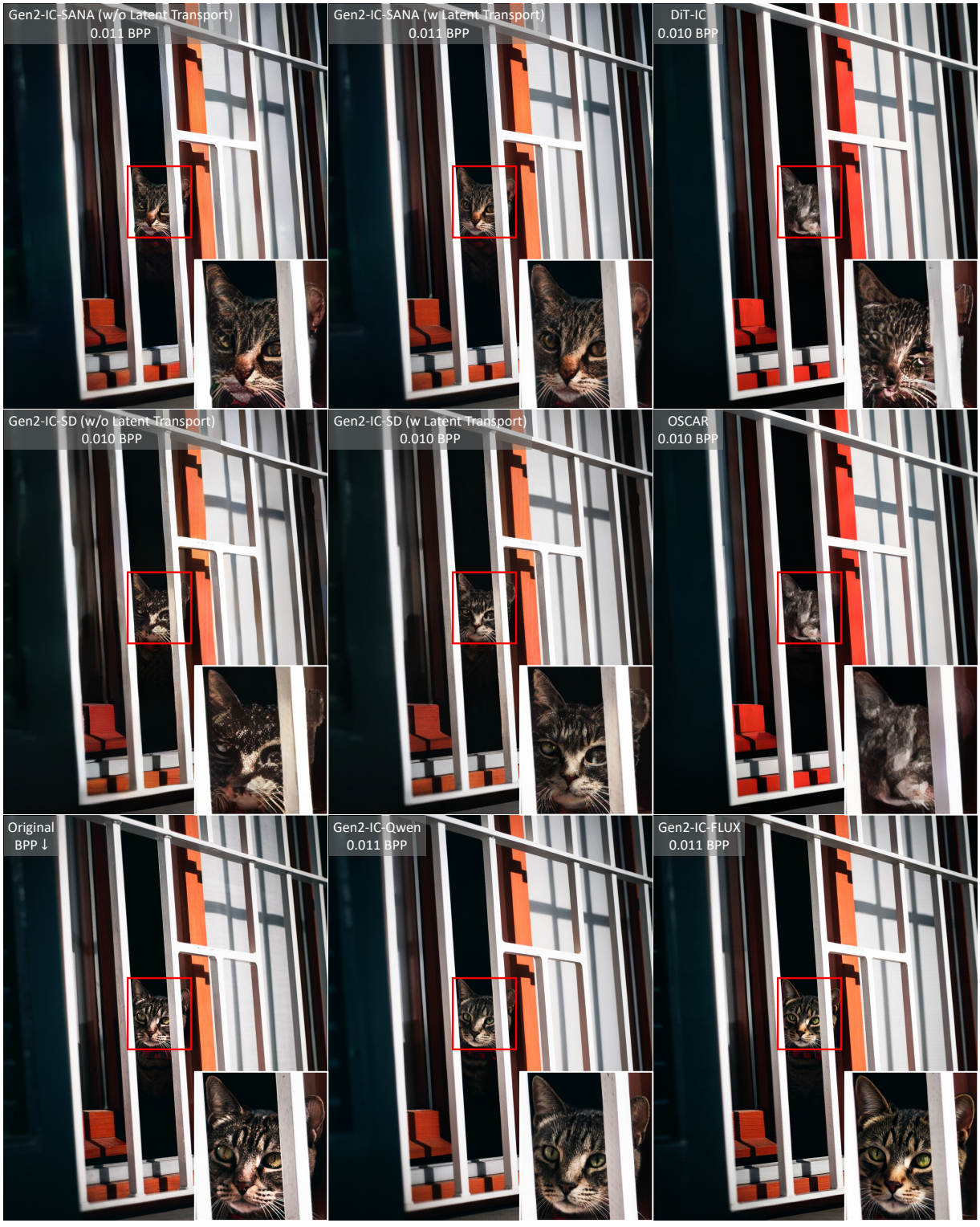


Figure S14: Matched-rate component and baseline comparison. Insets highlight the effect of Latent Transport on the partially occluded cat.

Figure S15: Component comparison of Gen2-IC-Base (pixel codec), VAE reconstruction, and Latent Transport for face reconstruction.

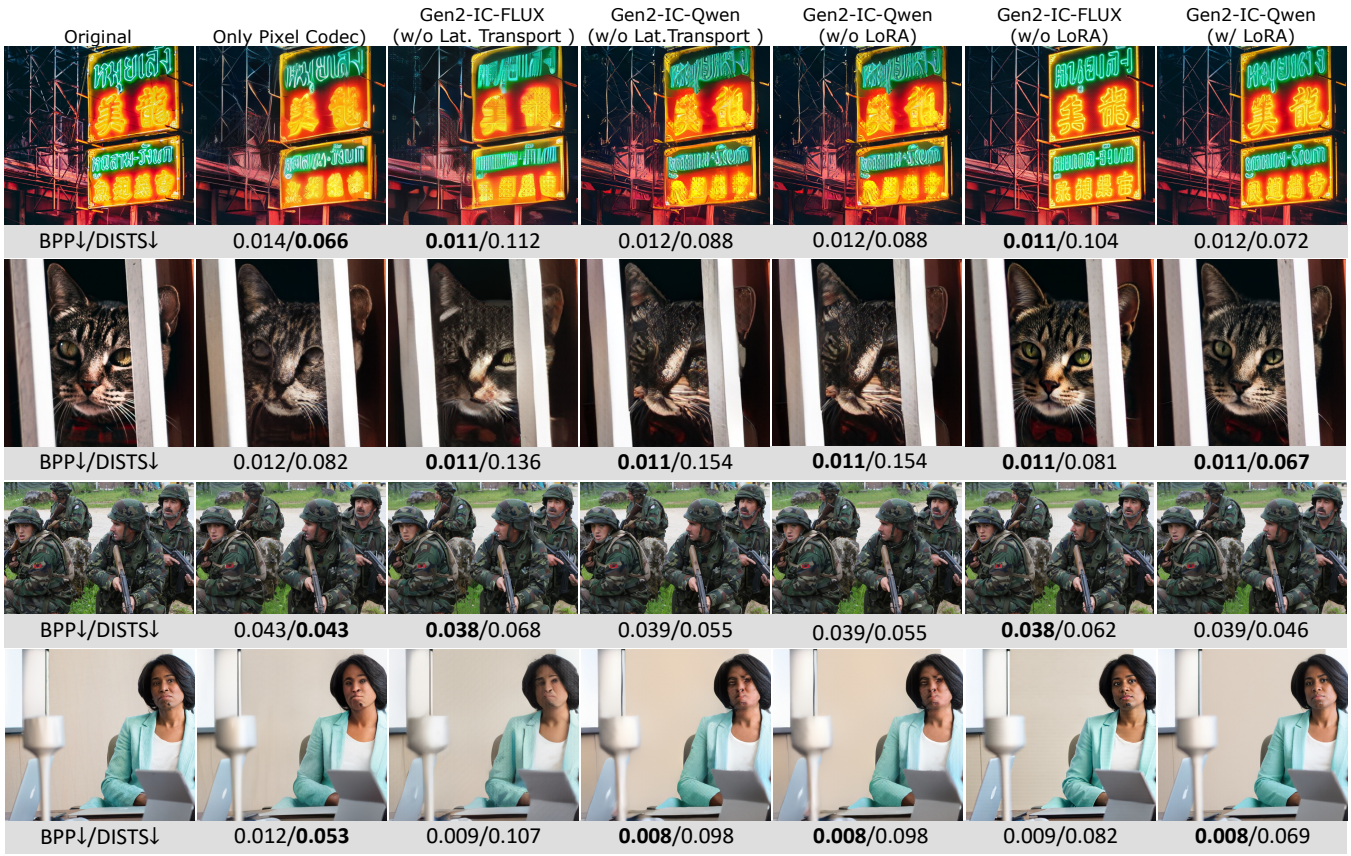


Figure S16: Component comparison across Gen2-IC-Base (pixel codec), VAE-only reconstruction, frozen-prior correction, and LoRA-adapted Latent Transport. Each row reports bpp/DISTS.